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The price of controversy: CEO Twitter activity and stock market performance☆

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ABSTRACT

This study examines how controversial CEO communication on social media influences stock market outcomes, providing a novel measure of organization-public relationship quality. Drawing on digital public relations and financial economics frameworks, we analyze controversial tweets sent by S&P 500 CEOs from 2009 to 2023. We identify controversial messages through two key indicators: the ratio of replies to retweets and sentiment analysis of public responses. Our findings reveal that controversial CEO tweets lead to immediate declines in firms' stock market value, with effects amplified for "celebrity" CEOs but partially buffered by high message visibility. These findings extend public relations theory by quantifying the reputational risks of executive social media communication and introducing public relations scholars to a novel, stock market-based framework for measuring both immediate and longer-term impacts of relationship damage.

1. Introduction

The evolution of digital platforms has fundamentally transformed how organizations build and maintain relationships with key stakeholders (Avidar & Roth-Cohen, 2023). Social media platforms, particularly Twitter,[1] have emerged as critical channels through which organizational leaders shape stakeholder relationships, manage corporate reputation, and cultivate public trust (Gómez & Chalmeta, 2011; Yim, 2019). While CEOs use these platforms to engage directly with stakeholders, this accessibility brings significant challenges. Though intended to strengthen relationships, controversial posts can trigger online "firestorms" that fundamentally damage organization-public relationships (Kim et al., 2021; Qu et al., 2023). Despite extensive research documenting the relationship-building potential of CEO social media engagement (Crowley et al., 2021; Elliott et al., 2018), we have limited understanding of how controversial or inflammatory CEO messages affect tangible measures of stakeholder relationship quality. Our study addresses this gap by examining how controversial CEO communication shapes measurable indicators of organization-public relationships.

Public relations scholarship has long sought reliable methods for measuring relationship quality between organizations and their publics. Recent digital public relations research suggests that stakeholder engagement metrics on social media can serve as meaningful indicators of relationship strength (Ji et al., 2019; Saxton & Waters, 2014). A key indicator of stakeholder disapproval occurs when a tweet receives more replies than retweets – a phenomenon known as being "ratioed." Because Twitter users are more likely to invest the additional effort to reply only when they find content objectionable, ratioed tweets are considered to serve as a valuable indicator of relationship strain caused by controversial social media messaging (Harris et al., 2023; Minot et al., 2021; Troy, 2022).

The implications of such strained relationships extend beyond immediate stakeholder reactions. We theorize that controversial CEO tweets – characterized by high reply-to-retweet ratios and negative sentiment – serve as public signals of deteriorating organization–public relationships (Hon & Grunig, 1999; Ledingham & Bruning, 2000). These signals can result in tangible reputational consequences, including stock price declines, as stakeholders withdraw support in ways that fundamentally alter the organization-public dynamic. We further expect these negative effects to be amplified when the CEO holds "celebrity" status

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[1] Throughout this study, we use the term "Twitter" rather than "X" to refer to the social media platform, as our analysis spans 2009–2023 when the platform was widely known as Twitter.

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(Freberg et al., 2011), due to increased public scrutiny, and moderated by tweet visibility (Nekrasov et al., 2021), with highly retweeted messages potentially diffusing or dampening the market response.

To test these hypotheses, we examine stock market reactions to S&P 500 CEO Twitter activity, analyzing both immediate market reactions to controversial tweets and extended effects surrounding quarterly earnings announcements. This dual approach provides tangible measures of how controversial CEO communication impacts organization-public relationships in both immediate and longer-term contexts.

This study advances public relations theory and practice in several ways (Avidar & Roth-Cohen, 2023). First, we extend relationship management theory by demonstrating how CEO social media activity directly shapes stakeholder relationships and organizational legitimacy. Second, by introducing stock market reactions (Uysal & Yang, 2013) as objective indicators of relationship damage, we provide a novel framework for measuring the tangible costs of controversial communication. Third, our findings offer empirical guidance for corporate leaders navigating the complex dynamics of social media engagement. Fourth, by centering on investors as a primary stakeholder group, we contribute to public relations scholarship by foregrounding a key but often overlooked group in communication research.

The paper proceeds as follows: We develop our theoretical framework linking CEO social media controversy to relationship outcomes in Section 2. We then detail our research methodology and present our findings in Sections 3 and 4. We conclude in Section 5 by discussing implications for relationship management theory and digital public relations practice.

2. Theory and hypothesis development

2.1. CEO communication, relationship management, and investor response

Public relations scholarship increasingly recognizes CEOs as influential relationship managers who shape stakeholder perceptions through public communication (Chandler, 2014; Freberg et al., 2011). On social media, CEOs function not merely as organizational spokespersons but as relationship cultivators whose messages can strengthen or strain stakeholder bonds (Cho et al., 2014; Ji et al., 2019). This role gained particular significance after 2013 when the SEC permitted CEOs to disclose market-relevant information through social media, elevating these platforms from general communication channels to vital investor relationship-building tools. In the digital age, the CEO has arguably become a "Chief Engagement Officer," particularly on platforms like Twitter, where the ability to cultivate immediate, authentic, and unfiltered relationships with publics introduces both opportunity and risk (Kim et al., 2016; Lee et al., 2025; Yue et al., 2019).

Given these dynamics, CEOs' social media communication is best understood through a relationship management lens. Relationship management theory posits that the goal of public relations is to build and maintain mutually beneficial relationships between organizations and their key publics (Broom et al., 1997; Hon & Grunig, 1999; Ledingham & Bruning, 2000). Hon and Grunig's (1999) organization-public relationship (OPR) theory refines this view by identifying four foundational dimensions of relationship quality: trust, satisfaction, control mutuality, and commitment. While controversial CEO tweets may affect multiple stakeholder groups, our analysis focuses on investors – a group actively cultivated by CEOs of publicly traded companies (Chandler, 2014) – whose responses are observable through stock market reactions. Stock market movements offer a readily quantifiable and timely measure of how these relationship dimensions are affected by CEO communication, as investor reactions aggregate into observable price changes (Pérez et al., 2020; Strauß et al., 2016; Uysal & Yang, 2013). Investor reactions are thus both financially meaningful and early signals of relationship strain that may foreshadow broader shifts in public sentiment (Pérez et al., 2020; Strauß et al., 2016).

2.2. Social media controversy and its relational and market consequences

CEO tweets are not merely informational updates; they are relational acts that stakeholders interpret as signals of trust, transparency, and intent. Messages from a CEO's personal account often carry greater symbolic weight and are perceived as more authentic than corporate communications (Elliott et al., 2018). Yet this authenticity can also amplify reputational risk – particularly when messages appear controversial or misaligned with stakeholder expectations. These communicative choices can strengthen stakeholder bonds or strain them, depending on how they are received and publicly interpreted.

Because these stakeholder responses play out in public view, platforms like Twitter allow researchers to observe relationship strain in real time. Metrics such as reply patterns, sentiment shifts, and engagement dynamics function as digital traces of organization–public relationship quality (Ji et al., 2019; Nekrasov et al., 2021; Saxton & Waters, 2014). These digital signals are consequential to capital markets because investors incorporate reputational cues into firm valuations, pricing relational breakdowns into stock market outcomes – a dynamic particularly useful for understanding relational damage when controversy arises.

We conceptualize social media controversies as public disagreements marked by conflicting viewpoints and polarized responses. These ongoing public disagreements are described as "the most heated debates on social media" (Garimella et al., 2018), frequently tied to broader social divisions (Kim & Ihm, 2020), and characterized by emotionally charged exchanges within echo-chamber network structures (Qiu et al., 2019). Importantly, social media controversy involves more than just heightened engagement; it signals negative sentiment and opposed perspectives that reflect underlying relationship strain and reputational risk. Building on this, we argue that controversial CEO tweets function as public signals of relational damage.

One visible marker of such controversy is being "ratioed," where a post receives more replies than retweets – a reversal of expected engagement patterns widely interpreted as a sign of disapproval or backlash. Scholars find ratioed messages reflect stakeholder resistance, challenge, or dissatisfaction – especially when replies express negative sentiment or signal breaches of social norms or expectations (Harris et al., 2023; Minot et al., 2021; Troy, 2022). These moments can escalate into online "firestorms" (Kim et al., 2021; Qu et al., 2023), generating intense scrutiny and reputational fallout. As such, being ratioed functions as a behavioral signal of contested communication and breakdowns in organization–public relationships.

The potential damage to organization-public relationships stemming from controversial CEO communication can translate into tangible negative outcomes for the organization. Specifically, controversial or inflammatory statements on social media can directly undermine stakeholders' trust in the CEO's and organization's integrity, decrease satisfaction with leadership, weaken commitment to the relationship, signal a lack of control mutuality, and shift perception from a communal relationship toward a more transactional view (Hon & Grunig, 1999; Ledingham & Bruning, 2000). This erosion of organization-public relationships can manifest through negative impacts on key OPR dimensions and may lead to broader stakeholder disengagement that alters the organization-public dynamic.

The implications extend beyond immediate stakeholder reactions. Evidence from the financial economics literature shows that firms' social media activities influence market outcomes (Bartov et al., 2018; Blankespoor et al., 2014), with negative stakeholder engagement affecting firm value (Alsabah, 2024). Building on these insights and grounded in the principles of relationship management theory and the OPR framework, in our first hypothesis we posit that controversial CEO tweets will damage organization-public relationships in ways that manifest in measurable stock market outcomes:

Hypothesis 1. Controversial CEO tweets are associated with lower stock market returns.

This hypothesis connects relationship management theory with financial outcomes, suggesting that breaches in CEO-stakeholder relationships – as indicated by controversial tweets – result in quantifiable organizational damage.

2.3. Star CEOs and message visibility: amplifying and buffering effects

Building on our core hypothesis about controversial CEO tweets, we now examine how different aspects of visibility may condition these relationships. We propose that *visibility* – both of the CEO and of the message itself – plays a crucial role in how controversial communication affects market outcomes. These visibility factors may function as either amplifiers or buffers in the relationship between CEO controversy and stakeholder reactions.

We first consider how CEO visibility, particularly in the case of "star" CEOs, may amplify negative reactions to controversial content. Drawing from relationship management theory, the actions and communications of highly visible leaders like "star" CEOs (Koh, 2011) carry greater weight in shaping stakeholder perceptions and organization-public relationship quality (Chandler, 2014; Elliott et al., 2018; Luo et al., 2014). On social media, these celebrity CEOs often function as "influencers" who shape the attitudes and perceptions of online communities (Freberg et al., 2011), with their symbolic leadership role reflected in firm performance (Bui et al., 2022). From an OPR perspective, star CEOs embody a greater share of the organization's identity, values, and relational commitments (Hon & Grunig, 1999). When these highly visible figures violate stakeholder trust – a core dimension of relationship quality – the perceived breach is amplified. Xu et al. (2024) highlight that public perception of corporate social advocacy often hinges on the alignment between a corporation's stance and audience political ideology, with high-profile CEOs potentially intensifying controversy and damaging stakeholder perceptions of legitimacy. In line with these arguments, our second hypothesis is:

Hypothesis 2. The negative relationship between controversial CEO tweets and stock market returns will be stronger for "star" CEOs.

Previous literature shows that retweet counts serve as an indicator of investor attention (Nekrasov et al., 2021). This message visibility changes information processing by stakeholders and may alter market reactions to controversial tweets. From a relationship management perspective, visibility adds complexity by expanding the conversation beyond the initial controversy (Ledingham & Bruning, 2000), potentially creating opportunities for control mutuality through broader stakeholder participation in shaping CEO communications' meaning and impact. As controversial messages circulate widely, they encounter diverse publics beyond initial negative reactors. This broader exposure may diffuse backlash intensity and provide opportunities for clarification, engagement, or reputational repair. Visibility, then, may attenuate relational damage by shifting the communication dynamic from reactive outrage to participatory dialogue (Lee et al., 2025; Olinski & Szamrowski, 2020). In line with these arguments, our third hypothesis is the following:

Hypothesis 3:. The negative relationship between controversial CEO tweets and stock market returns will be weaker for highly visible messages.

These hypotheses connect our primary relationship management framework to established findings about CEO visibility and message amplification on social media, allowing us to test whether digital relationship dynamics vary based on leader profile and content reach.

3. Research design and data

3.1. Event study design

To examine how controversial CEO communication affects organization-public relationships, we employ *event study* methodology from financial economics (Alsabah, 2024; Blankespoor et al., 2014; Nekrasov et al., 2021; Uysal & Yang, 2013). This approach analyzes how specific events impact firm value by measuring abnormal stock returns around event dates, as stock prices reflect new information, including potential damage to stakeholder relationships.

The key insight of event studies is that by examining stock returns in a narrow window around an event while controlling for broader market movements, researchers can isolate the event's impact from other factors. If returns during this window differ significantly from expectations based on normal market movements (generating "abnormal" returns), that difference can be attributed to the event being studied. In our context, negative abnormal returns following controversial CEO tweets would indicate that stakeholders consider the controversial communication to have damaged organization-public relationships in ways that affect firm value.

We conduct two complementary event studies. First, we consider the 569 individual *controversial tweets* as the "events" to capture immediate investor reactions to controversial messaging. Second, we analyze CEOs' level of controversy over extended 90-day periods preceding 1657 quarterly earnings announcement events (the standard "event" in financial economics studies); this allows us to assess whether sustained controversial behavior creates lasting relationship damage that manifests during key organizational moments.

This dual approach allows us to test each of our hypotheses at both the tweet and earnings announcement levels. The tweet-level analysis captures immediate stakeholder reactions to specific controversial messages, while the earnings announcement-level analysis reveals how accumulated controversial behavior influences investor perceptions during key corporate disclosure events. Together, these complementary approaches provide a more complete picture of how controversial CEO communication affects organization-public relationships across different temporal contexts.

3.2. Dependent variable: cumulative abnormal returns

For both tweet-level and earnings announcement-level tests, we estimate the following regression model:

$$CAR_{it} = \beta_0 + \beta_1 \times Controversy_{ijt} + Firm\ controls_{it} + Industry\ FE + Year\ FE + \varepsilon_{it} \quad (1)$$

CAR_{it} measures the cumulative abnormal return for firm *i* over the three-day window (*t-1* to $t+1$) surrounding each tweet/earnings announcement event *t* as described above. *CAR* is the most common measure of stock market response to external events in the financial economics literature; it measures the extent to which a stock's price responds "abnormally" to an event by comparing its performance (increase or decrease in price) to what would normally be expected based on past performance and overall market trends.[2] If *CAR* is positive, it suggests investors reacted positively to the event, while a negative *CAR* indicates a negative investor reaction.

3.3. Independent variables

3.3.1. Controversial tweeting

To identify controversial CEO tweeting, we employ a two-stage process combining engagement metrics with sentiment analysis. We began by accessing the Twitter API using custom Python code to download tweets (along with associated counts of retweets, likes, and replies and all reply text) sent from 2009 to 2023 by a list of S&P 500 CEO Twitter handles that we hand collected in 2020.[3]

We then calculated the reply-to-retweet ratio for each tweet, identifying those where replies exceeded retweets. Initial analysis revealed, surprisingly, that this metric alone was insufficient for capturing controversy, as a significant number of highly-replied tweets reflected positive engagement.[4] Therefore, we added a second criterion: sentiment analysis of replies.

We thus define a tweet as "controversial" when it both receives more replies than retweets and generates predominantly negative stakeholder responses, an approach supported by studies showing ratioed tweets typically reflect audience disapproval (Harris et al., 2023; Minot et al., 2021; Troy, 2022). This dual-criteria approach ensures our operationalization captures both the engagement pattern and negative sentiment necessary to indicate reputational strain.

To code reply sentiment, we employed RoBERTa (Rosenthal et al., 2017), a state-of-the-art language model ideal for this context. We selected RoBERTa because it is specifically fine-tuned on Twitter data, enabling it to effectively capture the nuanced and informal language patterns typical of the platform (Rosenthal et al., 2017). For our analysis, we utilized the RoBERTa implementation available through the Hugging Face Transformers library.

The model was used to classify each reply to a CEO's tweet as positive, negative, or neutral. This classification process was guided by the sentiment analysis training dataset from TweetEval, a comprehensive and peer-reviewed benchmark for evaluating language models on various Twitter-specific tasks (Barbieri et al., 2020).

The RoBERTa model demonstrates strong performance on this task, achieving an accuracy of 83 % on the TweetEval benchmark that significantly outperforms traditional lexicon-based approaches (Barbieri et al., 2020). It shows balanced performance across categories, with F1-scores of 0.81 for both negative and neutral classes, precision of 0.76 (negative) and 0.80 (neutral), and recall of 0.86 (negative) and 0.82 (neutral). In our study's dataset, this model classified 22.7 % of all replies as negative, 22.4 % as positive, and 54.9 % as neutral.

We then proceeded to operationalize our controversy variables. For our tweet-level tests, we first identified 569 controversial tweets, or those that receive more replies than retweets and generate predominantly negative stakeholder responses. For these tests our key variable, *Controversy,* is the reply-to-retweet ratio. In essence, this measure will allow us to see, for controversial tweets, whether the *degree* of controversy influences investor reactions.

Our earnings announcement tests, in turn, allow us to test whether investors react differently to CEOs who are more or less controversial over extended periods. For these tests we operationalize the following metric for each CEO, measured over the 90 days preceding each firm's quarterly earnings announcements: *Proportion Controversial* (the proportion of each CEO's tweets that are controversial) captures the frequency of controversial tweets in the pre-announcement period.[5]

3.3.2. Star CEO

To identify "celebrity" CEOs for Hypothesis 2, we follow prior research (Drake et al., 2012; Sabherwal & Uddin, 2019) by using Google search volume index (SVI) data. CEOs in the top quartile of SVI scores across our sample are assigned values of "1" on *Star CEO*, otherwise 0. This method is grounded in previous work that uses SVI to proxy for both executive visibility and symbolic prominence.

3.3.3. Message visibility

To test Hypothesis 3 regarding the moderating effects of message visibility, we created a binary *High Attention* measure based on retweet counts. For tweet-level tests, tweets in the highest retweet quartile were coded as 1, otherwise 0. For earnings announcement tests, CEOs in the highest quartile of average retweet counts during the 90-day pre-announcement period received scores of 1, otherwise 0. This operationalization aligns with previous research showing that retweet counts serve as a reliable indicator of investor attention and message reach (Nekrasov et al., 2021).

3.4. Control variables

To control for factors that might influence stock returns independently of controversy, we include standard control variables from the

[2] *CAR* represents the difference between actual and expected stock returns during the event window, where expected returns are estimated using a market-adjusted model based on the 100 days preceding the event. We calculate abnormal returns for days t-1, t, and t + 1 as the difference between actual and expected returns, then sum these values to obtain cumulative abnormal returns. This approach inherently controls for broader market movements during the event window by filtering out market-wide effects that would influence all stocks similarly. The inclusion of industry and year fixed effects in our regression models further accounts for industry-specific events and time-period effects. This methodological approach follows established practices in financial economics for isolating event-specific effects from broader market phenomena (Blankespoor et al., 2014; Nekrasov et al., 2021). As noted later, we also conducted robustness checks with longer event windows to ensure our results are not sensitive to the specific window length chosen.

[3] We started downloading CEO tweets in 2020 and thus have all tweets sent by the CEOs from 2020 to 2023 and, given the API limit of 3200 tweets per user, we could get tweets going back to as far as 2009 depending on how frequently a CEO tweeted.

[4] For example, when AMD CEO Lisa Su announces new products, her tweets often generate high reply counts but predominantly positive sentiment, reflecting healthy stakeholder engagement rather than controversy.

[5] For example, when analyzing T-Mobile's Q3 2022 earnings announcement (October 27, 2022), we examine CEO Mike Sievert's tweets over the 90-day period preceding the announcement (i.e., July 29, 2022 to October 26, 2022), calculating the proportion of tweets that were controversial.

capital markets literature (e.g., Blankespoor et al., 2014; Nekrasov et al., 2021): *Firm Size,* the log of total assets; *Book-to-Market,* the ratio of book value to market value; *Return on Assets,* net income divided by total assets; *Leverage*, total debt divided by total equity; and *Analyst Following*, the number of financial analysts following the firm. For the earnings announcement tests, we also include *Earnings Surprise*, which captures the difference between the company's reported earnings and analysts' expected earnings. All tests also include industry and year fixed effects to control for unobserved heterogeneity across industries and time. Definitions for all variables are included in Appendix A.

4. Analysis of findings

4.1. Overview of CEO controversial tweeting

Of the 80,000 + tweets in our initial sample, approximately 569 (less than 1 %) met our criteria for controversial messages – those receiving both high reply-to-retweet ratios and predominantly negative stakeholder sentiment. This relatively low proportion suggests that while digital reputation crises are not common, they represent significant departures from normal organization-public relationship patterns.

Fig. 1 presents examples of controversial and non-controversial CEO tweets. The non-controversial tweets, by Jim Fitterling (Dow Inc.) and Lisa Su (AMD), demonstrate higher engagement through likes and retweets than replies, indicating they are not "ratioed." In contrast, the controversial tweets, by Marc Benioff (Salesforce) and Mike Sievert (T-Mobile), have a higher number of replies than retweets – that is, they are "ratioed" – and the majority of replies are negative.

Fig. 2 provides a broader perspective on CEO communication patterns, displaying the average reply-to-retweet ratios across CEOs. This visualization reveals substantial variation in CEO social media engagement, with some leaders consistently generating more apparent stakeholder tension and controversy.

Fig. 3 examines relationship dynamics over time, tracking controversial versus non-controversial communication patterns for one CEO (T-Mobile's Mike Sievert) during June 2021. This granular view shows how controversial messages often occur alongside more positive interactions, suggesting the importance of considering both individual communications and broader engagement patterns.

4.2. Descriptive statistics

Table 1 presents descriptive statistics. Panel A highlights tweet-level variables, including *CAR* (0.002, SD = 0.031) and *Controversy* (mean = 4.196, SD = 10.511), indicating substantial variability in controversy levels across controversial tweets. Firm-level controls such as *Firm Size* (mean = 11.556) and *Book-to-Market* (mean = 0.349) show typical characteristics of large public firms. Panel B focuses on earnings announcement-level variables, where *Proportion Controversial* averages 0.006, highlighting the small fraction of tweets that are controversial.

4.3. Tests of Hypothesis 1: tweet-level and earnings announcement-level market reactions

Table 2 provides results for our primary hypothesis tests at both the tweet level (column 1) and the earnings announcement level (column 2). Looking first at the tweet-level tests, the coefficient for *Controversy* is negative and significant ($\beta = -0.0002$, $p < 0.05$), indicating that higher levels of controversy are associated with significant declines in investor returns. Control variables reveal that abnormal returns are significantly higher for smaller firms, firms with lower return on assets, those that are more leveraged, and those with higher analyst following.[6]

Looking at the earnings announcement-level tests, we see that *Proportion Controversial* is negatively associated with *CAR*.[7] These findings suggest that controversy, particularly during key company announcements, may undermine market optimism. Collectively, the results in Table 2 support Hypothesis 1.

4.4. Tests of Hypothesis 2: moderating role of star CEOs

Table 3 shows results from tests interacting our *Star CEO* variable with our measures of controversy at both the tweet level (column 1) and earnings announcement level (column 2). In both tests the main effect on controversy is significantly negative, while the coefficient on *Star CEO* is positive but not significant. Our main coefficient of interest, however, is on the interaction term. In the tweet-level test, the interaction between *Star CEO* and *Controversy* is significantly negative ($\beta = -0.003$, $p < 0.01$), suggesting that controversy around high-profile CEOs exacerbates negative market reactions. For earnings announcements, *Proportion Controversial* × *Star CEO* also shows a significant negative coefficient ($\beta = -0.80$, $p < 0.01$), reinforcing this amplifying effect in high-visibility contexts. In brief, when "star" CEOs post controversial messages, it intensifies the negative impact on the stock price. These findings provide strong support for Hypothesis 2, confirming that when "star" CEOs post controversial messages, it intensifies the negative impact on stock price.

4.5. Tests of Hypothesis 3: moderating role of message visibility

The results in Table 4 show mixed support for Hypothesis 3 regarding message visibility effects. In the tweet-level tests in column 1, the coefficient on *Controversy* × *High Attention* is positive and significant, lending support for our hypothesis that high message visibility mitigates some of the negative effects of controversy. However, for earnings announcements, no significant interaction effects involving high attention were observed, suggesting that the buffering effect of message visibility may be limited to immediate market reactions rather than extending to longer-term evaluation contexts.

5. Discussion and conclusions

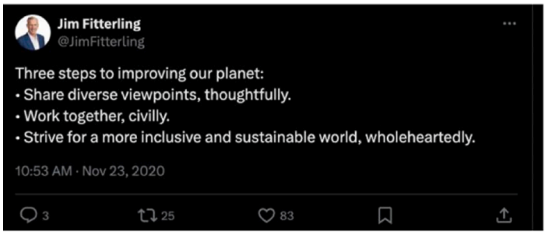
5.1. Contributions to theory and practice

Drawing on a comprehensive dataset of CEO tweets and market reactions from 2009 to 2023, we find strong evidence that controversial CEO communication on social media damages organizational value, with the effects intensified for high-profile leaders but partially buffered by broad message visibility. These findings offer several important implications for public relations theory and practice.

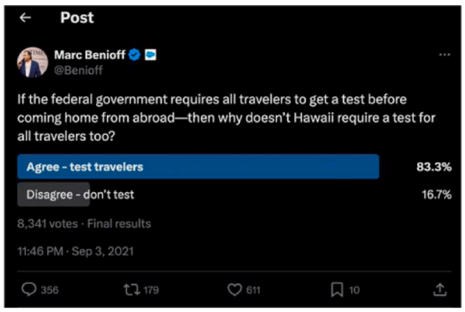
[6] To evaluate potential delayed investor reactions, we conducted robustness tests using extended event windows of [-5, + 5] and [-10, + 10] days. These analyses confirmed that the controversy variable maintains its negative association with CAR and achieves statistical significance over longer timeframes, suggesting markets gradually incorporate reputational signals rather than responding immediately to controversial CEO communications.

[7] We also ran tests using an alternative measure of controversy: *Average Controversy,* measured as the average reply-to-retweet ratio for each CEO in the 90-day period preceding each earnings announcement. The same result obtains as with *Proportion Controversial.* We also ran alternative analyses where we interact *Proportion Controversial* with our our measure of *Earnings Surprise* in order to show how the impact of controversy varies depending on the magnitude of unexpected earnings results. We find significant interaction effects. Specifically, the interaction of *Proportion Controversial* with *Earnings Surprise* is significantly negative ($\beta = -0.008$, $p < 0.01$), highlighting how controversy can dampen the positive effects of earnings surprises.

Non-Controversial CEO Tweets ***Controversial CEO Tweets***



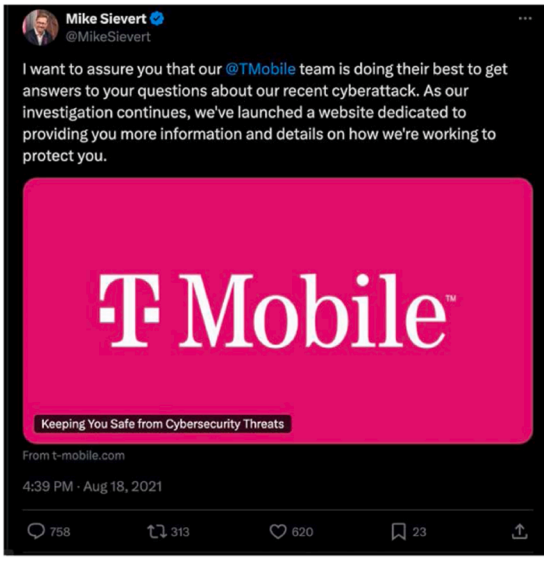
Non-controversial tweet with low attention



Controversial tweet – more replies than retweets (in addition, most replies are negative)



Non-controversial tweet with high attention



Controversial tweet – more replies than retweets (in addition, most replies are negative)

Fig. 1. Examples of Controversial and Non-Controversial Tweets. *Note:* Controversial tweets are those that are "ratioed," or have more replies than retweets, and where the majority of replies are negative.

First, we extend relationship management theory by demonstrating that controversial CEO messaging triggers measurable stock market penalties. By developing new metrics for identifying controversial communication and linking these to market reactions, we provide a novel framework for quantifying the tangible costs of damaged organization-public relationships. In so doing, we enrich Hon and Grunig's (1999) OPR framework by illustrating how its core dimensions play out on social media and translate into market impacts. While traditional relationship quality assessment relies on surveys, our approach captures relationship strain through digital indicators and connects them directly to financial outcomes. In addition, by foregrounding investors as a primary public, we show how PR research can better engage this vital – yet often overlooked – stakeholder group. Our results on CEO visibility and message reach further suggest boundary conditions for OPR theory: relationship effects vary based on communicator prominence and message diffusion. These extensions enhance existing models by showing how digital communication can reflect and shape stakeholder perceptions in ways that are immediate, measurable, and economically meaningful. This interdisciplinary approach bridges public relations and financial economics, deepening understanding of social media risks while expanding the conceptual toolkit available to PR researchers. Our work paves the way for future studies exploring the economic implications of digital communication strategies.

Second, our findings reveal how different forms of visibility shape relationship dynamics. Our finding that *CEO visibility* amplifies negative reactions while *message visibility* can buffer them suggests a complex interplay between leadership presence, stakeholder engagement, and relationship outcomes. This demonstrates how different forms of visibility distinctly affect organization-public relationships.

Third, we make a key methodological contribution through our measure of controversy. We find that just looking at "ratioed" tweets is insufficient; many of the ratioed messages in our sample were not controversial and instead reflected positive stakeholder engagement. For this very reason, we argue that the sentiment in public replies should also be considered. Our finding that the reply-to-retweet ratio, combined with sentiment analysis, predicts financial outcomes offers scholars a new measurement framework for assessing organizational messaging and digital relationship health.

Fourth, our study advances public relations methodology by demonstrating how regression-based event study analysis can systematically assess communication events' influence on stock market reactions. This approach, common in financial economics studies (e.g., Blankespoor et al., 2014; Nekrasov et al., 2021), offers considerable advantages to public relations scholars for isolating and quantifying the



Fig. 2. Average Reply-to-Retweet Ratio for S&P 500 CEOs. Figure presents a dot plot showing the average reply-to-retweet ratio for CEOs across the sample, offering a clear overview of how often tweets by different CEOs are "ratioed"—a key indicator of controversy. Each dot represents a CEO, with the x-axis showing their average reply-to-retweet ratio. Higher ratios indicate CEOs whose tweets are more likely to receive more replies than retweets, aligning with the study's definition of "controversial" tweets as those that provoke significant public backlash or negative sentiment. In contrast, lower ratios suggest CEOs whose tweets are less likely to elicit contentious discussions or disapproval.

impact of specific communication events – such as controversial tweets or crisis responses – from other factors affecting the organization. By analyzing controversial CEO tweets – an increasingly salient and understudied form of executive digital communication – we introduce individual-level messaging as a new type of event in PR stock market-based research. Our study captures both immediate market responses to specific tweets and sustained controversy effects surrounding earnings announcements, offering a broader, more nuanced framework for understanding digital relationship damage. This dual approach expands PR research beyond traditional organizational-level events like activist campaigns, CSR disclosures, or paracrises (Selaković et al., 2020; Uysal & Yang, 2013). Additionally, our large, diverse sample of S&P 500 firms enhances generalizability compared to studies focused on single organizations or regional markets (Pérez et al., 2020; Strauß et al., 2016; Uysal & Yang, 2013). This methodology offers rich opportunities for future research across multiple PR domains. By analyzing abnormal stock returns around key events – whether controversial tweets, natural disasters, or PR crises – researchers can isolate and quantify the impact of specific communication strategies. The granular, time-stamped nature of market data also allows for tracking how stakeholder reactions evolve over time, opening new possibilities for measuring organization-public relationships in ways that complement existing approaches.

Table 1
Descriptive statistics.

Panel A: Variables for tweet-level tests

	N	Mean	SD	Min	Max
CAR	569	0.002	0.031	−0.099	0.131
Controversy	569	4.196	10.511	1.029	134
Firm Size	569	11.556	1.093	9.092	13.045
Book-to-Market	569	0.349	0.307	−0.15	1.439
Return on Assets	569	0.069	0.064	−0.089	0.27
Analyst Following	569	21.466	6.103	7	46
Leverage	569	1.184	7.123	−9.998	82.074
Star CEO	569	0.07	0.256	0	1
High Attention	569	0.216	0.412	0	1

Table presents descriptive statistics for 569 tweet-level observations. Variables are defined in Appendix A

Panel B: Variables for earnings announcement-level tests

	N	Mean	SD	Min	Max
CAR	1657	0	0.067	−.342	0.526
Proportion Controversial	1657	0.006	0.053	0	1
Earnings Surprise	1657	0.667	9.748	−90.5	199
Firm Size	1657	9.772	1.459	6.879	13.045
Book-to-Market	1657	0.214	1.559	−44.101	1.684
Return on Assets	1657	0.073	0.081	−0.404	0.283
Analyst Following	1657	18.491	8.806	1	49
Leverage	1657	0.818	6.193	−60.829	82.074
Star CEO	1657	0.014	0.120	0	1
High Attention	1657	0.138	0.345	0	1

Table presents descriptive statistics for 1657 earnings announcement-level observations. Variables are defined in Appendix A.

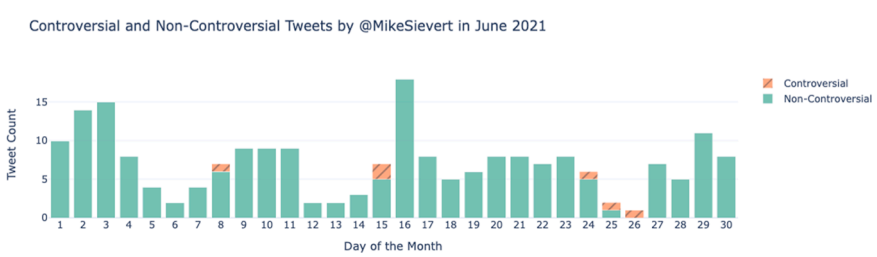


Fig. 3. Controversial and Non-Controversial Tweets Over Time for One CEO. Figure shows counts of controversial and non-controversial tweets sent in June 2021 by T-Mobile CEO Mike Sievert.

Table 2
Hypothesis 1 – Tweet-Level and Earnings Announcement-Level Tests.

	(1) DV=CAR (Tweet-Level)	(2) DV=CAR (Earnings Announcement-Level)
Controversy	−0.0002*	
	(0.0001)	
Proportion Controversial		−0.05^{+}
		(0.03)
Earnings Surprise		0.0004*
		(0.0002)
Firm Size	−0.03**	−0.002
	(0.005)	(0.002)
Book-to-Market	−0.007	−0.007
	(0.004)	(0.01)
Return on Assets	−0.09*	0.02
	(0.04)	(0.04)
Analysts Following	0.002**	0.0001
	(0.0004)	(0.0004)
Leverage	0.0003*	−0.0001
	(0.0001)	(0.0001)
Year and Industry Fixed Effects	YES	YES
Constant	0.30**	0.02
	(0.05)	(0.02)
Observations	569	1657
Adjusted R^2	0.060	0.006

Table presents results of event-study regressions of equation (1) at both the tweet level (column 1) and the earnings announcement level (column 2). For column (1), the "events" are the 569 controversial tweets, while for column (2) they are the 1, 657 quarterly earnings announcements. The independent variable in column (1) is the reply-to-retweet ratio for the controversial tweet, while in column (2) it is the proportion of controversial tweets sent by the CEO over the 90-day pre-earnings announcement period. For both tests the dependent variable *CAR* is the cumulative abnormal returns for S&P500 firms surrounding the respective event windows [-1 + 1]. Control variables are as defined in Appendix A. Standard errors are clustered at the firm level and are shown in parentheses. + p < 0.10, * p < 0.05, ** p < 0.01

Table 3
Conditional Effects of "Star" CEOs (H2) – Tweet-Level and Earnings Announcement-Level Tests.

	(1) DV=CAR (Tweet-Level)	(2) DV=CAR (Earnings Announcement-Level)
Controversy	−0.0002*	
	(0.00008)	
Proportion Controversial		−0.03*
		(0.01)
Star CEO	0.01	0.009
	(0.007)	(0.01)
Controversy × Star CEO	−0.003**	
	(0.0007)	
Proportion Controversial × Star CEO		−0.80**
		(0.06)
Earnings Surprise		0.0004*
		(0.0002)
Firm Size	−0.03**	−0.002
	(0.008)	(0.002)
Book-to-Market	−0.005	−0.006
	(0.01)	(0.01)
Return on Assets	−0.09*	0.02
	(0.04)	(0.04)
Analysts Following	0.0003*	0.00008
	(0.0001)	(0.0004)
Leverage	0.002**	−0.00007
	(0.0005)	(0.0001)
Year and Industry Fixed Effects	YES	YES
Constant	0.30**	0.02
	(0.08)	(0.02)
Observations	569	1657
Adjusted R^2	0.061	0.014

Table presents results from modified equation (1) regressions with an added interaction variable, *Star CEO*, which assigns values of 1 to CEOs in the highest quartile of the average score of Google's search volume index, otherwise 0. Column (1) presents the tweet-level test and column (2) shows the earnings announcement-level test. Control variables are as defined in Appendix A. Standard errors are clustered at the firm level and are shown in parentheses. + p < 0.10, * p < 0.05, ** p < 0.01

For practitioners, this study highlights the complex dynamics of organization-public relations in digital environments, particularly how controversial CEO communication can impact relationships and, ultimately, organizational value. While prior research underscores the benefits of CEO visibility (Crowley et al., 2021; Elliott et al., 2018), our findings emphasize the potential costs of controversy. Our findings particularly underscore the need for strategic oversight of executive social media presence. While blanket restrictions may be counterproductive – as high message visibility can sometimes mitigate negative reactions – organizations need sophisticated approaches to digital communication that balance engagement opportunities with risk management. For PR practitioners, our results emphasize the strategic importance of executive social media management and offer empirical guidance for protecting stakeholder relationships in digital environments.

5.2. Limitations and future research directions

While our study provides robust evidence linking controversial CEO communication to relationship outcomes, some limitations warrant consideration. Our focus on publicly traded companies means our findings may not fully generalize to other organizational contexts. Additionally, while market reactions provide an objective measure of relationship damage, they cannot capture all dimensions of organization-public relationships. Moreover, while we use a state-of-the-art sentiment analysis model (RoBERTa), we acknowledge that the absence of human-coded validation limits our ability to fully detect context-specific sentiment. Finally, an important limitation of our event study approach is the potential influence of concurrent market-wide or industry-specific events that may confound the observed relationship between controversial CEO tweets and abnormal stock returns. Although we control for general market trends using the S&P 500 index in calculating market-adjusted returns, macro-economic announcements (e.g., Federal Reserve policy changes), geopolitical developments, or sector-level shocks occurring near the event window may independently influence stock price movement. As a result, the attribution of market reaction solely to CEO communication should be interpreted with caution. Future research could address this limitation by applying intraday event analysis, incorporating economic calendar controls, or using natural experiments that better isolate CEO-driven events from other information shocks.

Despite these limitations, we believe our study makes significant contributions to understanding how executive communication shapes organization-public relationships in digital environments and points the way to several promising avenues of study. Future research could explore additional dimensions of CEO tweets, such as the tone, content, and timing, to provide a more comprehensive understanding of their market impact.

Table 4
Conditional Effects of Attention (H3) – Tweet-Level and Earnings Announcement-Level Tests.

	(1) DV=CAR (Tweet-Level)	(2) DV=CAR (Earnings Announcement-Level)
Controversy	−0.0002*	
	(0.00008)	
Proportion Controversial		−0.04
		(0.03)
High Attention	−0.0003	0.01*
	(0.004)	(0.005)
Controversy × High Attention	0.0008*	
	(0.0004)	
Proportion Controversial × High Attention		−0.04
		(0.04)
Earnings Surprise		0.0004*
		(0.0002)
Firm Size	−0.03**	−0.004*
	(0.005)	(0.002)
Book-to-Market	−0.006	−0.008
	(0.004)	(0.01)
Return on Assets	−0.09*	0.01
	(0.04)	(0.03)
Analysts Following	0.002**	0.0001
	(0.0005)	(0.0004)
Leverage	0.0003*	−0.00002
	(0.0001)	(0.0001)
Year and Industry Fixed Effects	YES	YES
Constant	0.3**	0.04*
	(0.05)	(0.02)
Observations	569	1657
Adjusted R^2	0.057	0.008

Table presents results from modified equation (1) regressions with an added interaction variable, *High Attention,* which assigns values of 1 to tweets (tweet-level test)/CEOs (earnings announcement-level test) in the highest quartile of number of retweets, otherwise 0. Column (1) presents the tweet-level test and column (2) shows the earnings announcement-level test. Control variables are as defined in Appendix A. Standard errors are clustered at the firm level and are shown in parentheses. + $p < 0.10$, * $p < 0.05$, ** $p < 0.01$

Future research could also explore how different crisis response strategies affect stock market reactions following controversial CEO tweets. Prior studies have examined the impact of CEO apologies (Koehn & Goranova, 2018; Zdaniuk & Chhinzer, 2019) and explanations (excuses, justifications, denials) on market outcomes, finding that sincerity and responsibility acknowledgment influence investor perceptions. The specific context of CEO-initiated social media controversies could reveal whether these findings translate to Twitter's fast-paced environment. Although our study focuses on the impact of controversial tweets themselves rather than subsequent responses, investigating how clarification, different apology types, strategic silence, or even self-mocking strategies (Kim et al., 2016) affect stock prices could provide valuable insights into how the timing, content, and presence of post-crisis communication shape market and stakeholder reactions.

Future studies could also investigate how audience demographics (e.g., political affiliation, consumer loyalty) moderate stakeholder reactions to controversial CEO tweets, and whether variations across industries influence the market impact of executive social media activity. These extensions would help identify contexts in which controversial communication poses greater or lesser risks to organizational value.

Another promising extension of this work would be to build on our sentiment analysis approach by incorporating human-coded validation of reply sentiment. This would help assess model accuracy and capture emotionally complex expressions – such as sarcasm, irony, or context-dependent responses – that machine learning models may overlook.

Our findings underscore the complex interplay between executive social media activity and organization-public relations in the digital age. While social media provides executives with valuable channels for direct stakeholder engagement, it also exposes firms to controversy-driven risks including weakened relationships, reputational damage, and market volatility. Organizations that master the delicate balance between leveraging executive voice and mitigating controversy will be best positioned to thrive in an increasingly digital future.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used *Claude* to make suggestions for improving readability. After using this tool, the authors reviewed and edited the content and take full responsibility for the content of the publication.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A

Appendix A: Variable DefinitionsVariable Name	Description
Dependent Variable	
CAR	Cumulative abnormal returns [t−1, t + 1], with expected returns based on a market-adjusted model calculated over the [−100, 0] pre-event window. For the tweet-level tests, the "event" is the controversial tweet, while for the earnings announcement-level tests the event is the earnings announcement date.
Independent Variables	
Controversy	Tweet-level measure. For each controversial tweet, this is measured as the reply-to-retweet ratio. Controversial tweets are identified as those that have more negative replies than retweets, and at least one reweet.
Proportion Controversial	Earnings announcement-level measure, operationalized as the number of controversial tweets divided by total number of tweets sent by each CEO in the 90-day period preceding each earnings announcement.
CEO/Tweet Characteristics	
Star CEO	Coded as 1 for CEOs in the highest quartile of the average score of Google's search volume index.
High Attention	For the tweet-level tests, coded as 1 for tweets that are in the highest quartile of retweet counts, otherwise 0. For the earnings announcement-level tests, coded as 1 for CEOs in the highest quartile of total number of retweets over the 90-day pre-event period, otherwise 0.
Control Variables	
Earnings Surprise	I/B/E/S reported actual quarterly earnings per share (EPS) less the latest I/B/E/S consensus analyst quarterly EPS forecast just prior to the earnings announcement, scaled by the standard deviation of estimates for the quarter.
Firm Size	Natural logarithm of total assets.

(*continued on next page*)

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Appendix A: Variable DefinitionsVariable Name	Description
Book-to-Market	The ratio of book to market value.
Leverage	The ratio of total debt to total equity.
Return on Assets	The ratio of net income to total assets.
Analyst Following	Number of financial analysts following the firm at the end of year.

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